

Haochuan Zhang

MS in Computer Science — Oregon State University

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SUMMARY

Applied AI researcher building physics-informed models for human body dynamics, with hands-on experience in robotics simulation (IsaacGym, MuJoCo), real-time 3D human perception, and multimodal sensor fusion.

Led the full pipeline from data generation (AMASS → PHC → IsaacGym 1024-env rollout) to State Space Model training and physics-aware evaluation, targeting sim-to-real transfer for embodied control.

Strong systems background: CUDA/Triton kernel-level optimization, cloud GPU inference (AWS/GCP), and end-to-end deployment of ML pipelines to production.

EDUCATION

Oregon State University

MS in Computer Science, GPA: 3.97

Corvallis, OR

Sep 2024 – Jun 2026 (expected)

University of Kansas

BS in Computer Science

Lawrence, KS

Aug 2017 – May 2022

RESEARCH EXPERIENCE

Graduate Research Assistant — PhysMamba: Physics-Informed SSM for Human Dynamics

Oregon State University, Computer Vision Group

Corvallis, OR

Jan 2025 – Present

- Exploited the structural analogy between Mamba3 SSM state equations $x(t+1)=Ax(t)+Bu(t)$ and articulated rigid-body dynamics to build a learned physics simulator for humanoid bodies, achieving lowest rollout divergence on 500-step autoregressive prediction vs GRU and MLP baselines at matched 3M parameter scale.
- Constructed a scalable data pipeline (AMASS → PHC humanoid controller → IsaacGym 1024-parallel-env rollout) generating 500K+ state-action-next_state transitions; applied LARP-style action perturbation protocol (controller isolation + external force) to validate causal action→state learning over trajectory memorization.

Graduate Research Assistant — 3D Human Mesh Recovery

Oregon State University, Computer Vision Group

Corvallis, OR

Jan 2025 – Present

- Benchmarked SAM 3D-Body (DINOv3-H+, 840M params), 4D-Humans, and HMR2.0b on in-the-wild tennis video; identified systematic temporal drift in occluded joints via PA-MPJPE and per-joint error analysis, motivating the physics-informed SSM approach.
- Implemented sliding-window smoothing and recurrent temporal aggregation following GLAMR and WHAM; reduced per-frame jitter and established baselines that PhysMamba's sequential state prediction further improved upon.

Research Fellow — Multimodal Robotics Perception (NEST-ARTERD)

University of Tokyo & Japan Atomic Energy Agency (JAEA)

Tokyo & Fukushima, Japan

Mar 2025 – Aug 2025

- Fused RGB, depth, LiDAR, and radiation sensors on mobile robotic platforms for autonomous 3D mapping in nuclear decommissioning environments; developed calibration and temporal alignment pipelines across heterogeneous sensor modalities.
- Validated perception robustness under real-world degradation (occlusion, dynamic lighting, radiation interference), producing reproducible experimental workflows adopted by the JAEA robotics team.

PROJECTS

Retrieval-Augmented Generation for Code and Patent Search | Milvus, sentence-transformers

2025 – 2026

- Combined tree-sitter AST-aware chunking with hybrid BM25+dense retrieval (Reciprocal Rank Fusion) and BGE cross-encoder reranking; achieved Recall@5 of 0.89, +23% over naive split. Applied NV-Retriever's positive-aware hard negative mining and CLP loss to fine-tune code embeddings (Recall@5: 0.74→0.82).
- Extended the hybrid retrieval approach to patent prior-art search with 5 parallel recall channels and sentence-transformer reranking; designed a 4-agent LangGraph evaluation pipeline for structured novelty assessment.

Diffusion Transformers and Latent Generative Models | PyTorch, HuggingFace diffusers

2025

- Trained DDPM on CIFAR-10 (FID~5) and reproduced Latent Diffusion downsampling ablation ($f=4/8/16$) on CelebA-HQ; applied Min-SNR- γ weighting to resolve inter-timestep gradient conflicts, reproducing the $3.4\times$ convergence speedup.

- Replaced the UNet backbone with a DiT (Peebles & Xie) Transformer architecture on class-conditional ImageNet 256²; studied the scaling behavior of DiT-S/B/L/XL and the effect of adaLN-Zero conditioning. Extended to Rectified Flow (SD3) and Consistency Models for few-step sampling.

3D Gaussian Splatting for Real-Time Scene Reconstruction | COLMAP, PyTorch, CUDA 2025

- Reproduced 3D Gaussian Splatting (Kerbl et al. SIGGRAPH 2023) on custom-captured multi-view scenes; achieved >100 FPS real-time rendering at comparable PSNR/SSIM to NeRF while reducing training time from hours to minutes via differentiable tile-based rasterization.
- Investigated adaptive density control (clone/split/prune) and spherical harmonics degree for view-dependent appearance; compared with Instant-NGP and Mip-NeRF 360 on Tanks&Temples and indoor scenes.

VLM/VLA for Robotic Manipulation | PyTorch, Open X-Embodiment 2025

- Studied RT-2 (Brohan et al.) and OpenVLA (Kim et al. 2024) architectures mapping vision-language inputs to robot action tokens; fine-tuned OpenVLA-7B on domain-specific manipulation tasks using LoRA, evaluating action prediction accuracy on the Bridge V2 benchmark.
- Explored π_0 (Physical Intelligence, 2024) flow-matching action head for continuous control and compared with autoregressive action tokenization; analyzed the trade-off between action discretization granularity and manipulation success rate.

Personalized Conversational Agent | LLM, LoRA, RLHF/PPO, Milvus, RAG 2025

- Fine-tuned a large language model with LoRA on 200K+ real conversation records to capture individual speaking style; applied RLHF with PPO to align response tone with user preferences, achieving higher human-rated naturalness over the base model.
- Built a RAG-based conversation memory using Milvus vector database over the full chat history; achieved sub-50ms retrieval latency at scale, significantly outperforming SQL full-text search on long-tail conversational queries.

PUBLICATIONS

H. Zhang, et al., *PhysMamba: Physics-Informed State Space Model for Human Body Dynamics Prediction*, in preparation, targeting CVPR/ICCV 2026.

Y. Xie, J. Shang, Q. Yang, X. Qian, **H. Zhang**, X. Tang, *ARSA-UNet: Atrous residual network based on Structure-Adaptive model for retinal vessel segmentation*, Biomedical Signal Processing and Control, Elsevier, 2025.

TECHNICAL SKILLS

Languages: Python, C/C++, Objective-C, Bash, SQL, JavaScript/TypeScript

ML & Vision: PyTorch, Mamba/SSM, Transformers, SMPL/SMPL-H, SAM 3D-Body, 4D-Humans, HMR2.0b, YOLO

Robotics & Simulation: IsaacGym, MuJoCo, PHC (physics-based humanoid controller), AMASS

HPC & GPU: CUDA, Triton kernels, OpenMP, SIMD, MPI, SLURM

LLM/RAG: LangGraph, vector databases (Milvus), hybrid retrieval (BM25 + dense), sentence-transformers, VLM

Infrastructure: AWS (EC2, S3, Lambda), GCP Cloud Run, Docker, Kubernetes

PATENTS

Contributor, CN117743818A: Domain Adaptation Method for Motor Imagery EEG Signals (2024)